# Digital Electronics

Previous Year Questions • 1991–2024

## GATE ELECTRICAL ENGINEERING

**(EE)**

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

# Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Digital Electronics (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

## Meet your Authors


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- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience

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- AIR 19 — GATE (IN)
- AIR 66 — GATE (EE)
- 3+ years in teaching & the VLSI industry

> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

## How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

**Q3.24.7**

| | |
|---|---|
| **3** | Unit (chapter) number — Unit III of this book |
| **24** | GATE exam year — 2024 (two digits; 98 = 1998) |
| **7** | Question number within that unit and year |

So **Q3.24.7** is the 7$^{th}$ question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

## What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

## How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

| | |
|---|---|
| **Questions** | the real total for this unit. |
| **MCQ, MSQ, NAT** | that same total, split by question type. |
| **1M, 2M** | how many of those questions carry 1 or 2 marks. |
| **Descriptive** | questions worth more than 2 marks, or whose marks value isn't recorded. |
| **Avg Weightage** | the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one. |
| **Range** | the lowest and highest that same per-sitting value has been, with the year each occurred. |

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

### Worked example

**Unit Snapshot***

**Stream:** EC **Questions:** 45 **MCQ:** 30 **MSQ:** 3 **NAT:** 12
**1M:** 20 **2M:** 22 **Descriptive:** 3 **Avg Weightage** ~4M **Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

# Important Links & Resources

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

## Contributors

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UNIT

I

# Logic Gates and Boolean Algebra

## Topics

1. Introduction to Digital Systems
2. Basic Gates and Concept of Multivibrators Using NOT Gate
3. Universal and Arithmetic Gates
4. Implementing Boolean Expressions Using Universal Logic Gates
5. Timing Diagrams
6. Boolean Algebra, Venn Diagrams, and the Concept of Dual

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## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 14 | **MCQ:** 14 | **MSQ:** 0 | **NAT:** 0 |
| **1M:** 10 | **2M:** 4 | **Descriptive:** 0 | **Avg Weightage** ~2M | **Range:** 1M(2024)–2M(2026) |

**See preface for how Avg Weightage and Range are calculated.*

**Q1.26.1** MCQ 2026 2M

The digital circuit shown has 3 inputs ($x$, $y$, and $z$).

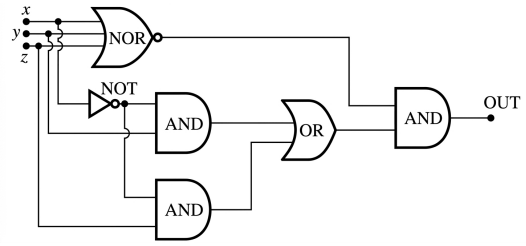


Fig. Q1.26.1

The simplified logical expression for the output (OUT) is

**A.** $\bar{x}\,\bar{y}\,\bar{z}$ **B.** 0
**C.** $\bar{x}(y+z)$ **D.** 1

**Q1.24.1** MCQ 2024 1M

Simplified form of the Boolean function

$$F(P,Q,R,S) = \bar{P}\bar{Q} + \bar{P}QS + P\bar{Q}\bar{R}\bar{S} + P\bar{Q}R\bar{S}$$

is

**A.** $\bar{P}S + \bar{Q}\bar{S}$

**B.** $\bar{P}\bar{Q} + \bar{Q}\bar{S}$

**C.** $\bar{P}\bar{Q} + R\bar{S}$

**D.** $P\bar{S} + Q\bar{R}$

**Q1.19.1** MCQ 2019 2M

In the circuit below, $X$ and $Y$ are digital inputs, and $Z$ is a digital output. The equivalent circuit is a

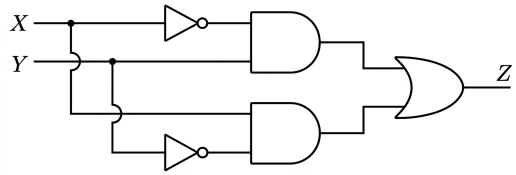


Fig. Q1.19.1

**A.** XOR gate **B.** XNOR gate
**C.** NAND gate **D.** NOR gate

**Q1.18.1** MCQ 2018 1M

In the logic circuit shown in the figure, $Y$ is given by

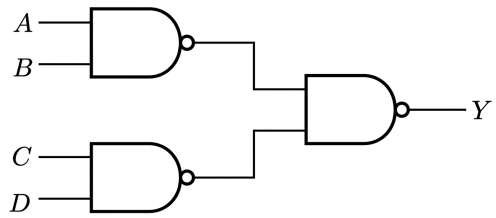


Fig. Q1.18.1

**A.** $Y = ABCD$

**B.** $Y = (A+B)(C+D)$

**C.** $Y = A + B + C + D$

**D.** $Y = AB + CD$

**Q1.17.1** MCQ 2017 1M

For a three-input logic circuit shown below, the output $Z$ can be expressed as

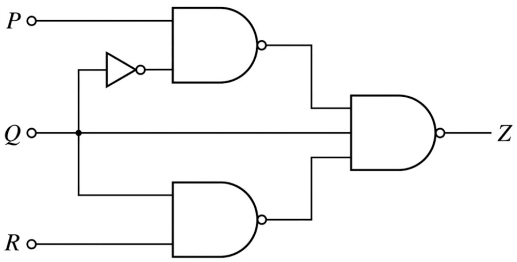


Fig. Q1.17.1

**A.** $Q + \bar{R}$ **B.** $P\bar{Q} + R$
**C.** $\bar{Q} + R$ **D.** $P + \bar{Q} + R$

**Q1.13.1** MCQ 2013 1M

A bulb in a staircase has two switches, one switch being at the ground floor and the other one at the first floor. The bulb can be turned ON and also can be turned OFF by any one of the switches irrespective of the state of the other switch.

The logic of switching of the bulb resembles

**A.** An AND gate

**B.** An OR gate

**C.** An XOR gate

**D.** A NAND gate

**Q1.11.1** MCQ 2011 1M

The output $Y$ of the logic circuit given below is

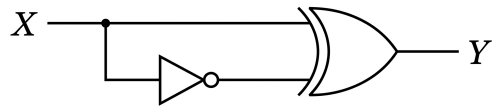


Fig. Q1.11.1

**A.** 1 **B.** 0
**C.** $X$ **D.** $\bar{X}$

**Q1.09.1** MCQ 2009 1M

The complete set of only those Logic Gates designated as Universal Gates is

**A.** NOT, OR and AND Gates

**B.** XNOR, NOR and NAND Gates

**C.** NOR and NAND Gates

**D.** XOR, NOR and NAND Gates

**Q1.07.1** MCQ 2007 1M

A, B, C and D are input bits and $Y$ is the output bit in the XOR gate circuit shown below. Which of the following statements about the sum $S$ of $A$, $B$, $C$, $D$ and $Y$ is correct?

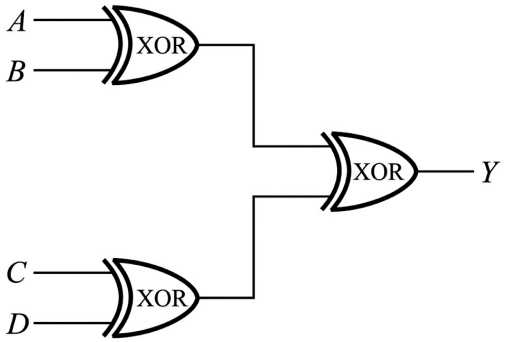


Fig. Q1.07.1

**A.** $S$ is always either zero or odd.

**B.** $S$ is always either zero or even.

**C.** $S = 1$ only if the sum of $A$, $B$, $C$ and $D$ is even.

**D.** $S = 1$ only if the sum of $A$, $B$, $C$ and $D$ is odd.

**Q1.02.1** MCQ 2002 2M

For the circuit shown in figure, the Boolean expression for the output $Y$ in terms of inputs $P$, $Q$, $R$ and $S$ is

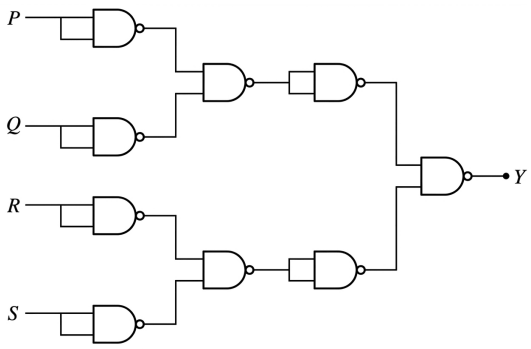


Fig. Q1.02.1

**A.** $\bar{P} + \bar{Q} + \bar{R} + \bar{S}$

**B.** $P + Q + R + S$

**C.** $(\bar{P} + \bar{Q})(\bar{R} + \bar{S})$

**D.** $(P + Q)(R + S)$

**Q1.01.1** MCQ 2001 1M

The output of a logic gate is "1" when all its inputs are at logic "0". The gate is either

**A.** A NAND or an EX-OR gate

**B.** A NOR or an EX-OR gate

**C.** An AND or an EX-NOR gate

**D.** A NOR or an EX-NOR gate

**Q1.99.1** MCQ 1999 2M

The logic function $f = \overline{(x \cdot \bar{y}) + (\bar{x} \cdot y)}$ is the same as

**A.** $f = (x + y)(\bar{x} + \bar{y})$

**B.** $f = \overline{(\bar{x} + \bar{y})(x + y)}$

**C.** $f = \overline{(x \cdot y)}(\bar{x} \cdot \bar{y})$

**D.** None of the above

**Q1.96.1** MCQ 1996 1M

The Boolean expression of the output of the logic circuit shown in figure is

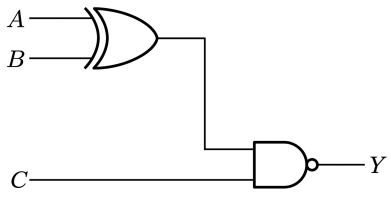


Fig. Q1.96.1

**A.** $Y = \bar{A}\bar{B} + AB + \bar{C}$

**B.** $Y = \bar{A}\bar{B} + AB + C$

**C.** $Y = \bar{A}B + \bar{A}B + C$

**D.** $Y = \bar{A}B + \bar{A}\bar{B} + \bar{C}$

**Q1.91.1** MCQ 1991 1M

For the combinational circuit shown in figure,

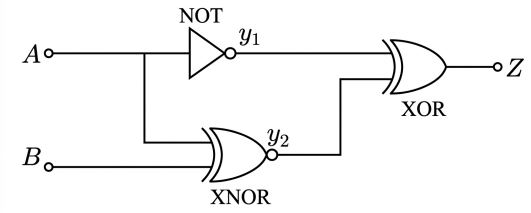


Fig. Q1.91.1

which of the following truth tables is correct?

(A)

| A | B | Z |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

(B)

| A | B | Z |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

(C)

| A | B | Z |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

(D)

| A | B | Z |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

Fig. Q1.91.1

**A.** (A) **B.** (B) **C.** (C) **D.** (D)

## Answer Key

| | | |
|---|---|---|
| Q1.26.1 B | Q1.24.1 A | Q1.19.1 A |
| Q1.18.1 D | Q1.17.1 C | Q1.13.1 C |
| Q1.11.1 A | Q1.09.1 C | Q1.07.1 B |
| Q1.02.1 B | Q1.01.1 D | Q1.99.1 B |
| Q1.96.1 A | Q1.91.1 A | |

UNIT

II

# Representation of Boolean Expressions and K-Maps

## Topics

1. Minterms and Maxterms
2. SOP and POS Representation
3. Karnaugh Maps (K-Maps)
4. Concept of Implicants

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## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 17 | **MCQ:** 17 | **MSQ:** 0 | **NAT:** 0 |
| **1M:** 7 | **2M:** 10 | **Descriptive:** 0 | **Avg Weightage** ~2M | **Range:** 1M(2015)–2M(2025) |

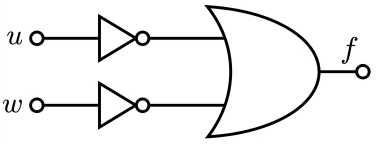
**See preface for how Avg Weightage and Range are calculated.*

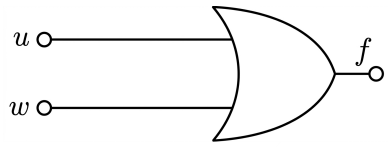
**Q2.25.1** MCQ 2025 2M

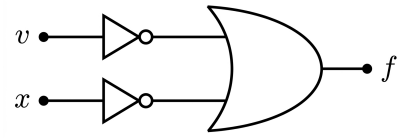
A Boolean function is given as

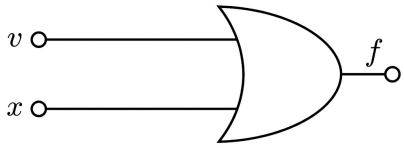
$$f = (\bar{u}+\bar{v}+\bar{w}+\bar{x})\cdot(\bar{u}+\bar{v}+\bar{w}+x)\cdot(\bar{u}+v+\bar{w}+\bar{x})\cdot(\bar{u}+v+\bar{w}+x)$$

The simplified form of this function is represented by

**A.**



**B.**



**C.**



**D.**



**Q2.19.1** MCQ 2019 2M

The output expression for the Karnaugh map shown below is

| PQ / RS | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 1 | 1 | 0 |
| 01 | 1 | 1 | 1 | 1 |
| 11 | 1 | 1 | 1 | 1 |
| 10 | 0 | 0 | 0 | 0 |

Fig. Q2.19.1

**A.** $QR + \bar{S}$
**B.** $Q\bar{R} + S$
**C.** $QR + S$
**D.** $Q\bar{R} + \bar{S}$

**Q2.18.1** MCQ 2018 2M

Digital input signals $A$, $B$, $C$ with $A$ as the MSB and $C$ as the LSB are used to realize the Boolean function

$$F = m_0 + m_2 + m_3 + m_5 + m_7,$$

where $m_i$ denotes the $i^{\text{th}}$ minterm. In addition, $F$ has a don't care for $m_1$. The simplified expression for $F$ is given by

**A.** $\bar{A}\bar{C} + \bar{B}C + AC$
**B.** $\bar{A} + C$
**C.** $\bar{C} + A$
**D.** $\bar{A}C + BC + A\bar{C}$

**Q2.17.1** MCQ 2017 1M

The Boolean expression

$$AB + A\bar{C} + BC$$

simplifies to

**A.** $BC + A\bar{C}$
**B.** $AB + A\bar{C} + B$
**C.** $AB + A\bar{C}$
**D.** $AB + BC$

**Q2.17.2** MCQ 2017 2M

The output expression for the Karnaugh map shown below is

| AB / CD | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 0 | 0 | 0 |
| 01 | 1 | 0 | 0 | 1 |
| 11 | 1 | 0 | 1 | 1 |
| 10 | 0 | 0 | 0 | 0 |

Fig. Q2.17.2

**A.** $B\bar{D} + BCD$
**B.** $B\bar{D} + AB$
**C.** $\bar{B}D + ABC$
**D.** $B\bar{D} + ABC$

**Q2.16.1** MCQ 2016 1M

The output expression for the Karnaugh map shown below is

| A / BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

Fig. Q2.16.1

**A.** $A + \bar{B}$ **B.** $A + \bar{C}$
**C.** $\bar{A} + \bar{C}$ **D.** $\bar{A} + C$

**Q2.16.2** MCQ 2016 2M

The Boolean expression

$$\overline{(a + \bar{b} + c + \bar{d}) + (b + \bar{c})}$$

simplifies to

**A.** 1 **B.** $\overline{ab}$
**C.** $ab$ **D.** 0

**Q2.15.1** MCQ 2015 1M

$f(A, B, C, D) = \Pi M(0, 1, 3, 4, 5, 7, 9, 11, 12, 13, 14, 15)$ is a maxterm representation of Boolean function $f(A, B, C, D)$ where $A$ is the MSB and $D$ is the LSB. The equivalent minimized representation of this function is

**A.** $(A + \bar{C} + D)(\bar{A} + B + D)$

**B.** $A\bar{C}D + \bar{A}BD$

**C.** $\bar{A}C\bar{D} + A\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D}$

**D.** $(B + \bar{C} + D)(A + \bar{B} + \bar{C} + D)(\bar{A} + B + C + D)$

**Q2.15.2** MCQ 2015 1M

Consider the following Sum of Products expression, $F$.

$$F = ABC + \bar{A}\bar{B}C + A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}\bar{C}$$

The equivalent Product of Sums expression is

**A.** $(A + \bar{B} + C)(\bar{A} + B + C)(\bar{A} + \bar{B} + C)$

**B.** $(A + \bar{B} + \bar{C})(A + B + C)(\bar{A} + \bar{B} + \bar{C})$

**C.** $(\bar{A} + B + \bar{C})(A + \bar{B} + \bar{C})(A + B + C)$

**D.** $(\bar{A} + \bar{B} + C)(A + \bar{B} + \bar{C})(A + B + C)$

**Q2.14.1** MCQ 2014 2M

The SOP (sum of products) form of a Boolean function is $\Sigma(0, 1, 3, 7, 11)$, where inputs are $A, B, C, D$ ($A$ is MSB and $D$ is LSB). The equivalent minimised expression of the function is

**A.** $(\bar{B} + C)(\bar{A} + C)(\bar{A} + \bar{B})(\bar{C} + D)$

**B.** $(\bar{B} + C)(\bar{A} + C)(\bar{A} + \bar{C})(\bar{C} + D)$

**C.** $(\bar{B} + C)(\bar{A} + C)(\bar{A} + \bar{C})(\bar{C} + \bar{D})$

**D.** $(\bar{B} + C)(A + \bar{B})(\bar{A} + \bar{B})(\bar{C} + D)$

**Q2.14.2** MCQ 2014 2M

Which of the following logic circuits is a realization of the function $F$ whose Karnaugh map is shown in the figure?

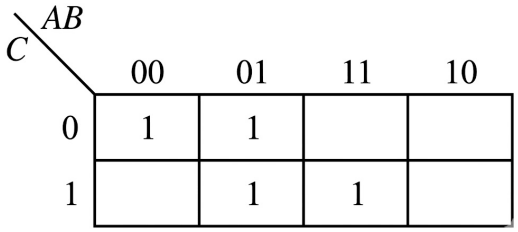
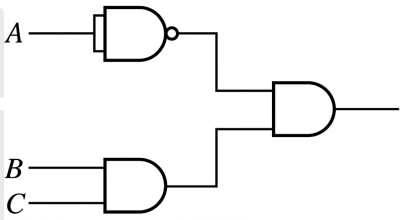
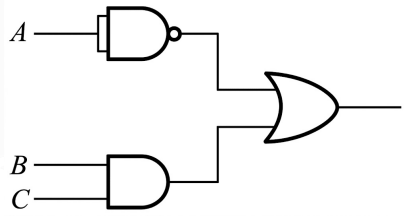
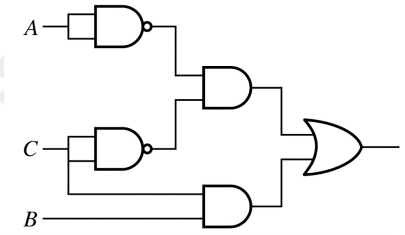
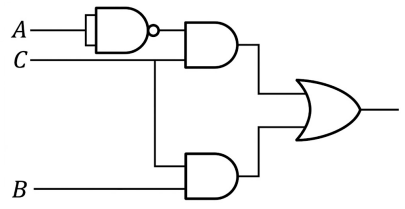



Fig. Q2.14.2

**A.**



**B.**



**C.**



**D.**



**Q2.12.1** MCQ 2012 1M

In the sum of products function

$$f(X, Y, Z) = \sum m(2, 3, 4, 5)$$

the prime implicants are

**A.** $\bar{X}Y,\ X\bar{Y}$

**B.** $\bar{X}Y,\ X\bar{Y}Z,\ X\bar{Y}\bar{Z}$

**C.** $\bar{X}Y\bar{Z},\ \bar{X}YZ,\ X\bar{Y}$

**D.** $\bar{X}Y\bar{Z},\ \bar{X}YZ,\ X\bar{Y}\bar{Z},\ X\bar{Y}Z$

## Common Data Questions

**The next 2 questions share the data below.**

Answer Questions Q2.10.1 and Q2.10.2 using the following data.

The following Karnaugh map represents a function $F$:

| $F$: $X$ \ $YZ$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 | 0 |

**Fig. Q2.12.1**

### Q2.10.1 MCQ 2010 1M

A minimised form of the function $F$ is

**A.** $F = \bar{X}Y + YZ$

**B.** $F = \bar{X}\bar{Y} + YZ$

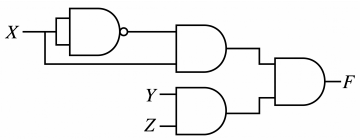
**C.** $F = \overline{XY} + Y\bar{Z}$

**D.** $F = \overline{XY} + \bar{Y}Z$

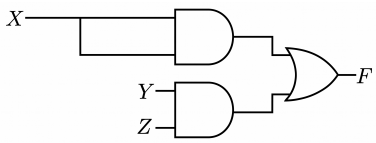
### Q2.10.2 MCQ 2010 1M

Which of the following circuits is a realisation of the above function $F$?

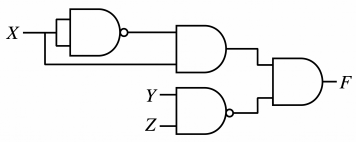
**A.**



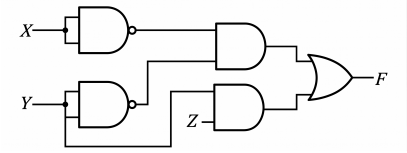

**B.**




**C.**




**D.**




### Q2.04.1 MCQ 2004 2M

The simplified form of the Boolean expression

$$Y = (\bar{A}BC + D)(\bar{A}D + B\bar{C})$$

can be written as

**A.** $\bar{A}D + B\bar{C}D$
**B.** $AD + B\bar{C}D$
**C.** $(\bar{A} + D)(B\bar{C} + \bar{D})$
**D.** $A\bar{D} + BC\bar{D}$

### Q2.03.1 MCQ 2003 2M

The Boolean expression

$$\bar{X}Y\bar{Z} + \bar{X}\bar{Y}Z + XY\bar{Z} + X\bar{Y}Z + XYZ$$

can be simplified to

**A.** $X\bar{Z} + \bar{X}Z + YZ$
**B.** $XY + \bar{Y}Z + Y\bar{Z}$
**C.** $\bar{X}Y + YZ + XZ$
**D.** $\overline{XY} + Y\bar{Z} + \bar{X}Z$

### Q2.00.1 MCQ 2000 2M

The minimal product-of-sums function described by the K-map given in figure is

| $C$ \ $AB$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 1 | 1 | $\phi$ | 0 |
| 1 | 0 | 0 | $\phi$ | 0 |

**Fig. Q2.00.1**

**A.** $\bar{A}\bar{C}$
**B.** $\bar{A} + \bar{C}$
**C.** $A + C$
**D.** $AC$

## Answer Key

| | | |
|---|---|---|
| Q2.25.1<br>A | Q2.19.1<br>B | Q2.18.1<br>B |
| Q2.17.1<br>A | Q2.17.2<br>D | Q2.16.1<br>B |
| Q2.16.2<br>D | Q2.15.1<br>C | Q2.15.2<br>A |
| Q2.14.1<br>A | Q2.14.2<br>C | Q2.12.1<br>A |
| Q2.10.1<br>B | Q2.10.2<br>D | Q2.04.1<br>A |
| Q2.03.1<br>B | Q2.00.1<br>A | |

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UNIT

III

# Number Systems

## Topics

1. Introduction to Number Systems
2. Arithmetic Operations on Numbers
3. Binary Codes and Their Operations
4. Signed Number Representations
5. Complement Arithmetic
6. Floating-Point Representation

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## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 2 | **MCQ:** 2 | **MSQ:** 0 | **NAT:** 0 |
| **1M:** 1 | **2M:** 1 | **Descriptive:** 0 | **Avg Weightage** ~1M | **Range:** 1M(2014)–1M(2014) |

**See preface for how Avg Weightage and Range are calculated.*

**Q3.14.1** MCQ 2014 2M

Which of the following is an invalid state in 8 4 2 1 Binary Coded Decimal counter?

**A.** 1000 **B.** 1001

**C.** 0011 **D.** 1100

**Q3.07.1** MCQ 2007 1M

The Octal equivalent of the HEX number $AB.CD$ is

**A.** 253.314

**B.** 253.632

**C.** 526.314

**D.** 526.632

## Answer Key

| Q3.14.1 | Q3.07.1 | |
|---|---|---|
| D | B | |

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UNIT

# IV

# Combinational Circuits

## Topics

1. Introduction to Combinational Circuits
2. Code Converters and 7-Segment Display
3. Parity Generator and Checker
4. Magnitude Comparators
5. Multiplexers (MUX) and Demultiplexers (DEMUX)
6. Implementing Logical Expressions Using MUX
7. Decoders, Encoders and Priority Encoders
8. Half Adder and Full Adder
9. Delay-Based Problems in Adders
10. Half Subtractor and Full Subtractor
11. Binary Parallel Adder with Delay-Based Problems
12. Look-Ahead Carry Adder
13. Binary Parallel Subtractor
14. Adder and Subtractor Circuits of Different Binary Codes
15. Binary Multiplier

## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 13 | **MCQ:** 12 | **MSQ:** 0 | **NAT:** 0 |
| **1M:** 6 | **2M:** 7 | **Descriptive:** 0 | **Avg Weightage** ~1M | **Range:** 1M(2024)–2M(2015) |

**See preface for how Avg Weightage and Range are calculated.*

**Q4.24.1** MCQ 2024 1M

To obtain the Boolean function $F(X, Y) = X\bar{Y} + \bar{X}$, the inputs $P$, $Q$, $R$, $S$ in the figure should be

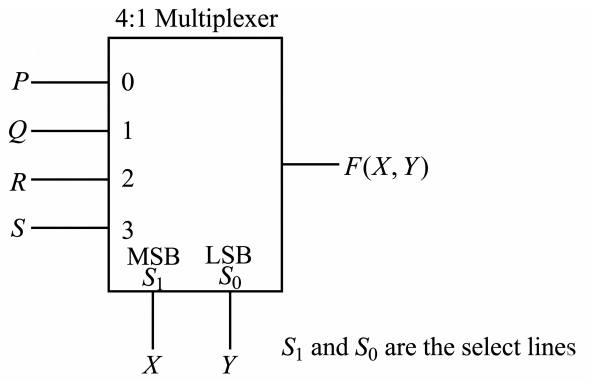


Fig. Q4.24.1

**A.** 1010 **B.** 1110
**C.** 0110 **D.** 0001

**Q4.16.1** MCQ 2016 1M

Consider the following circuit which uses a 2-to-1 multiplexer as shown in the figure below. The Boolean expression for output $F$ in terms of $A$ and $B$ is

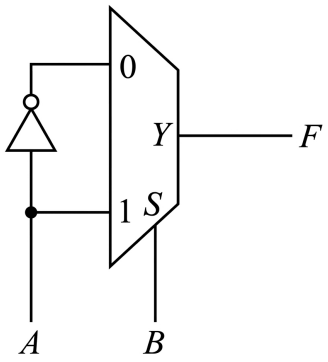


Fig. Q4.16.1

**A.** $A \oplus B$ **B.** $\overline{A + B}$
**C.** $A + B$ **D.** $\overline{A \oplus B}$

**Q4.15.1** MCQ 2015 1M

In the $4 \times 1$ multiplexer, the output $F$ is given by $F = A \oplus B$. Find the required input $I_3 I_2 I_1 I_0$.

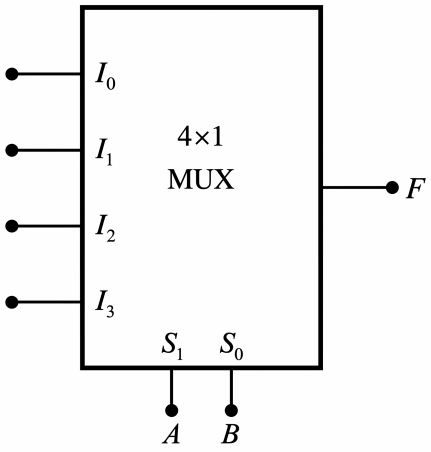


Fig. Q4.15.1

**A.** 1010 **B.** 0110
**C.** 1000 **D.** 1110

**Q4.15.2** MCQ 2015 2M

A Boolean function $f(A, B, C, D) = \Pi M(1, 5, 12, 15)$ is to be implemented using an $8 \times 1$ multiplexer ($A$ is MSB). The inputs $ABC$ are connected to the select input $S_2 S_1 S_0$ of the multiplexer respectively. Which one of the following options gives the correct inputs to pins 0, 1, 2, 3, 4, 5, 6, 7 in order?

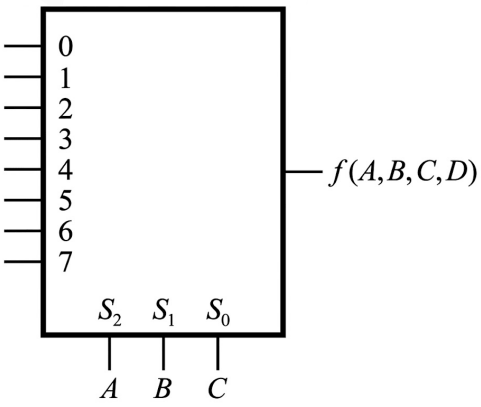


Fig. Q4.15.2

**A.** $D$, 0, $D$, 0, 0, 0, $\bar{D}$, $D$ **B.** $\bar{D}$, 1, $\bar{D}$, 1, 1, 1, $D$, $\bar{D}$
**C.** $D$, 1, $D$, 1, 1, 1, $\bar{D}$, $D$ **D.** $\bar{D}$, 0, $\bar{D}$, 0, 0, 0, $D$, $\bar{D}$

**Q4.14.1** MCQ 2014 2M

A 3-bit gray counter is used to control the output of the multiplexer as shown in the figure. The initial state of the counter is $000_2$. The output is pulled high. The output of the circuit follows the sequence.

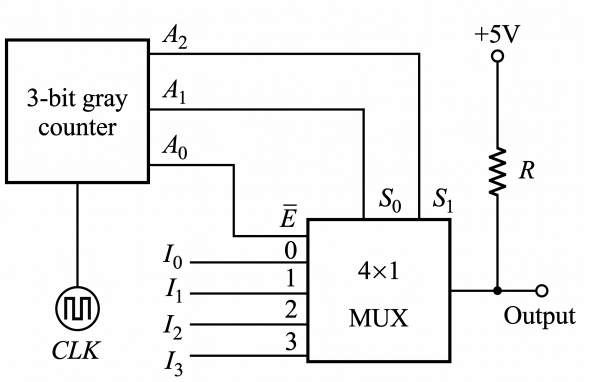


**Fig. Q4.14.1**

**A.** $I_0, 1, 1, I_1, I_3, 1, 1, I_2$

**B.** $I_0, 1, I_1, 1, I_2, 1, I_2, 1$

**C.** $1, I_0, 1, I_1, 1, I_2, 1, I_3, 1$

**D.** $I_0, I_1, I_2, I_3, I_0, I_1, I_2, I_3$

## Q4.12.1 MCQ 2012 1M

The output $Y$ of a 2-bit comparator is logic 1 whenever the 2-bit input $A$ is greater than the 2-bit input $B$. The number of combinations for which the output is logic 1, is

**A.** 4 **B.** 6
**C.** 8 **D.** 10

## Q4.08.1 MCQ 2008 2M

A 3-line to 8-line decoder, with active low outputs, is used to implement a 3-variable Boolean function as shown in the figure.

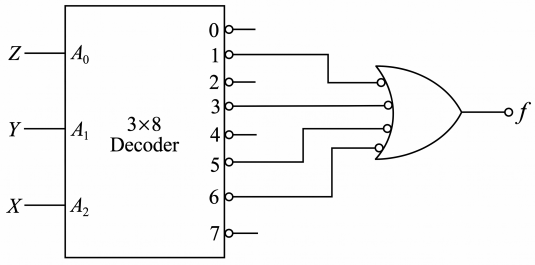


**Fig. Q4.08.1**

The simplified form of Boolean function $F(X, Y, Z)$ implemented in 'Product of Sum' form will be

**A.** $(X + Z) \cdot (\bar{X} + \bar{Y} + \bar{Z}) \cdot (Y + Z)$

**B.** $(\bar{X} + \bar{Z}) \cdot (X + Y + Z) \cdot (\bar{Y} + \bar{Z})$

**C.** $(\bar{X} + \bar{Y} + Z) \cdot (\bar{X} + Y + Z) \cdot (X + \bar{Y} + Z) \cdot (X + Y + \bar{Z})$

**D.** $(\bar{X} + \bar{Y} + Z) \cdot (\bar{X} + Y + Z) \cdot (X + \bar{Y} + Z) \cdot (X + \bar{Y} + \bar{Z})$

## Q4.06.1 MCQ 2006 2M

A $4 \times 1$ MUX is used to implement a 3-input Boolean function $F(A, B, C)$ as shown in the figure. The Boolean function $F(A, B, C)$ implemented is

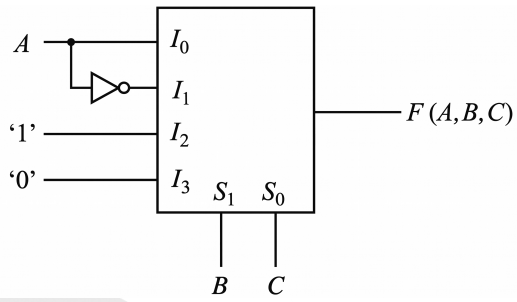


**Fig. Q4.06.1**

**A.** $F(A, B, C) = \sum m(1, 2, 4, 6)$

**B.** $F(A, B, C) = \sum m(1, 2, 6)$

**C.** $F(A, B, C) = \sum m(2, 4, 5, 6)$

**D.** $F(A, B, C) = \sum m(1, 5, 6)$

## Q4.04.1 MCQ 2004 2M

A digital circuit which compares two numbers $A_3\, A_2\, A_1\, A_0$, $B_3\, B_2\, B_1\, B_0$ is shown in the figure. To get output $Y = 0$, choose one pair of correct input numbers.

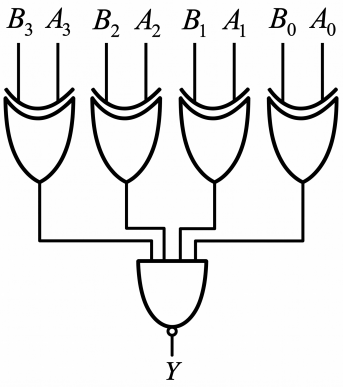


**Fig. Q4.04.1**

**A.** 1010, 1010 **B.** 0101, 0101
**C.** 0010, 0010 **D.** 1010, 0101

## Q4.03.1 MCQ 2003 1M

Figure shows a 4-to-1 MUX to be used to implement the sum $S$ of a 1-bit full adder with input bits $P$ and $Q$ and the carry input $C_{in}$. Which of the following combinations of inputs to $I_0$, $I_1$, $I_2$ and $I_3$ of the MUX will realise the sum $S$?

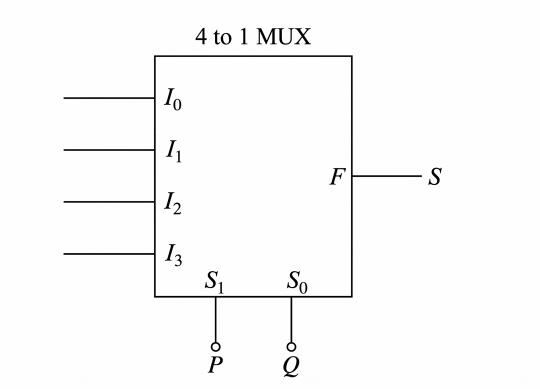


Fig. Q4.03.1

**A.** $I_0 = I_1 = C_{in};\ I_2 = I_3 = \bar{C}_{in}$

**B.** $I_0 = I_1 = \bar{C}_{in};\ I_2 = I_3 = C_{in}$

**C.** $I_0 = I_3 = C_{in};\ I_1 = I_2 = \bar{C}_{in}$

**D.** $I_0 = I_3 = \bar{C}_{in};\ I_1 = I_2 = C_{in}$

## Q4.01.1 MCQ 2001 1M

The output $F$ of the 4-to-1 MUX shown in the figure is

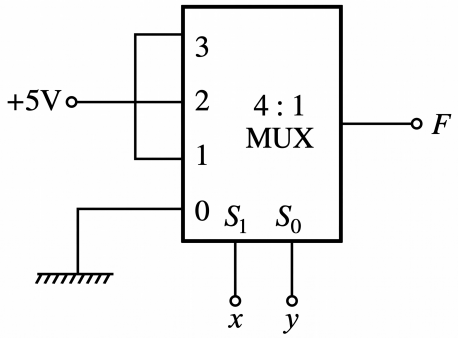


Fig. Q4.01.1

**A.** $\overline{xy} + x$

**B.** $x + y$

**C.** $\bar{x} + \bar{y}$

**D.** $xy + \bar{x}$

## Q4.99.1 Descriptive 1999 2M

The logic function

$$F = AC + ABD + ACD$$

is to be realized using an 8-to-1 multiplexer shown in the figure, using $A$, $C$ and $D$ as control inputs.

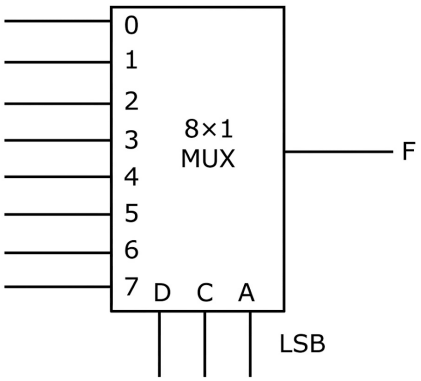


Fig. Q4.99.1

1. Indicate the inputs to be applied at the terminals 0 to 7.
2. Can the function be realized using a 4-to-1 multiplexer? State **YES** or **NO**.

## Q4.98.1 MCQ 1998 2M

Match the following:

| Logic | Function |
|---|---|
| I. $\bar{X} + \bar{Y}$ | P. Sum |
| II. $XY$ | Q. NAND |
| III. $\bar{X}\bar{Y}$ | R. Carry |
| | S. NOR |

**A.** I-Q, II-R, III-S

**B.** I-S, II-R, III-Q

**C.** I-Q, II-P, III-S

**D.** I-S, II-P, III-Q

## Answer Key

**Note:** * descriptive/evidence-based, no single definitive answer.

| | | |
|---|---|---|
| Q4.24.1<br>B | Q4.16.1<br>D | Q4.15.1<br>B |
| Q4.15.2<br>B | Q4.14.1<br>A | Q4.12.1<br>B |
| Q4.08.1<br>A | Q4.06.1<br>A | Q4.04.1<br>D |
| Q4.03.1<br>C | Q4.01.1<br>B | Q4.99.1<br>* |
| Q4.98.1<br>A | | |

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UNIT

# V

# Sequential Circuits

## Topics

1. Latches — SR, JK, D and T Types, Their Drawbacks and Gated Latches
2. Flip-Flops — Introduction, Types, Conversions, Race Around Condition and Master-Slave Configuration
3. Asynchronous Counters — Binary, Non-Binary (Up/Down) and Output Decoding
4. Synchronous Counters — Propagation Delay, Setup/Hold Time, Critical Path Delay, Clear and Preset Input
5. Shift Registers and Applications — Ring Counter, Johnson Counter and Cascading
6. FSM and Sequence Detectors — Design, Generation, Detection and Overlapping/Non-Overlapping Types
7. Static Timing Analysis — Setup/Hold Time, Clock Constraints, Critical Path and Timing Violations

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## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 34 | **MCQ:** 27 | **MSQ:** 0 | **NAT:** 7 |
| **1M:** 17 | **2M:** 14 | **Descriptive:** 3 | **Avg Weightage** ~2M | **Range:** 1M(2024)–3M(2021) |

**See preface for how Avg Weightage and Range are calculated.*

**Q5.24.1** MCQ 2024 1M

In the circuit, the present value of $Z$ is 1. Neglecting the delay in the combinational circuit, the values of $S$ and $Z$ respectively, after the application of the clock will be

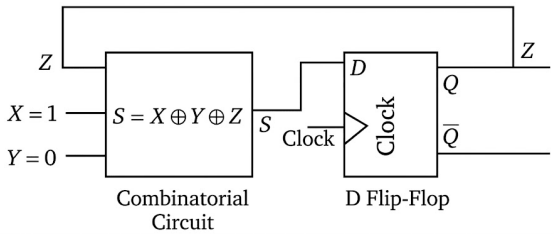



Fig. Q5.24.1

**A.** $S = 0,\ Z = 0$
**B.** $S = 0,\ Z = 1$
**C.** $S = 1,\ Z = 0$
**D.** $S = 1,\ Z = 1$

**Q5.23.1** NAT 2023 2M

Neglecting the delays due to the logic gates in the circuit shown in figure, the decimal equivalent of the binary sequence $[ABCD]$ of initial logic states, which will not change with clock, is

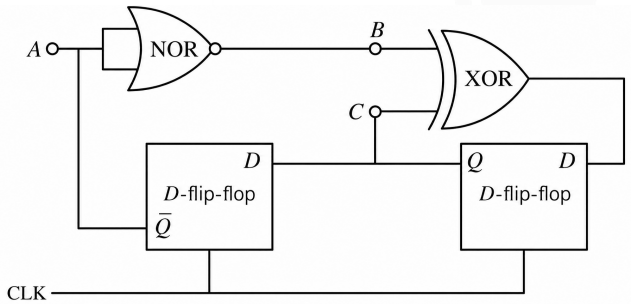



Fig. Q5.23.1

**Q5.22.1** NAT 2022 1M

A MOD 2 and a MOD 5 up-counter combined together results in a MOD _____ counter. (in integer)

**Q5.22.2** NAT 2022 1M

The maximum clock frequency in MHz of a 4-stage ripple counter, utilizing flip-flops, with each flip-flop having a propagation delay of 20 ns, is _____. (round off to one decimal place)

**Q5.21.1** NAT 2021 1M

A 16-bit synchronous binary up-counter is clocked with a frequency $f_{clk}$. The two most significant bits are OR-ed together to form an output $Y$. Measurements show that $Y$ is periodic and the duration for which $Y$ remains high in each period is 24 ms. The clock frequency $f_{clk}$ is _____ MHz. (Round off to 2 decimal places)

**Q5.21.2** MCQ 2021 2M

A counter is constructed with three $D$ flip-flops. The input-output pairs are named as $(D_0, Q_0)$, $(D_1, Q_1)$ and $(D_2, Q_2)$, where the subscript 0 denotes LSB. The output sequence is desired to be Gray-code sequence 000, 001, 011, 010, 110, 111, 101 and 100, repeating periodically. Note that the bits are listed in the $Q_2\,Q_1\,Q_0$ format. The combination logic expression for $D_1$ is

**A.** $Q_2\,Q_1\,Q_0$

**B.** $Q_2\,Q_0 + Q_1\bar{Q}_0$

**C.** $\bar{Q}_2\,Q_0 + Q_1\bar{Q}_0$

**D.** $\bar{Q}_2\,Q_1 + \bar{Q}_2\bar{Q}_1$

**Q5.20.1** MCQ 2020 1M

A sequence detector is designed to detect precisely 3 digital inputs, with overlapping sequences detectable. For the sequence $(1, 0, 1)$ and input data $(1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0)$, what is the output of this detector?

**A.** 1, 1, 0, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0

**B.** 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0

**C.** 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0

**D.** 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 0

**Q5.18.1** MCQ 2018 2M

Which one of the following statements is true about the digital circuit shown in the figure?

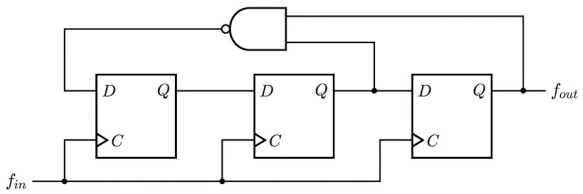



Fig. Q5.18.1

**A.** It can be used for dividing the input frequency by 3.

**B.** It can be used for dividing the input frequency by 5.

**C.** It can be used for dividing the input frequency by 7.

**D.** It cannot be reliably used as a frequency divider due to disjoint internal cycles.

### Q5.17.1 NAT 2017 2M

For the synchronous sequential circuit shown below, the output $Z$ is zero for the initial conditions $Q_A\,Q_B\,Q_C = Q'_A\,Q'_B\,Q'_C = 100$.

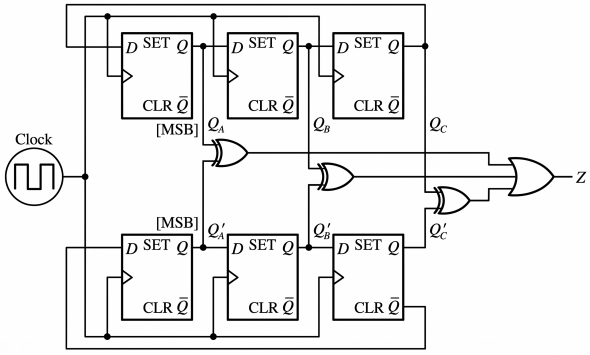



Fig. Q5.17.1

The minimum number of clock cycles after which the output $Z$ would again become zero is _____.

### Q5.16.1 MCQ 2016 2M

The current state $Q_A\,Q_B$ of a two $J$-$K$ flip-flop system is 00. Assume that the clock rise-time is much smaller than the delay of the $J$-$K$ flip-flop. The next state of the system is

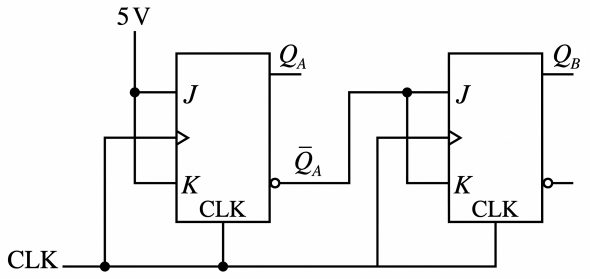



Fig. Q5.16.1

**A.** 00 **B.** 01
**C.** 11 **D.** 10

### Q5.15.1 MCQ 2015 2M

In the following sequential circuit, the initial state (before the first clock pulse) of the circuit is $Q_1\,Q_0 = 00$. The state $(Q_1\,Q_0)$, immediately after the 333rd clock pulse is

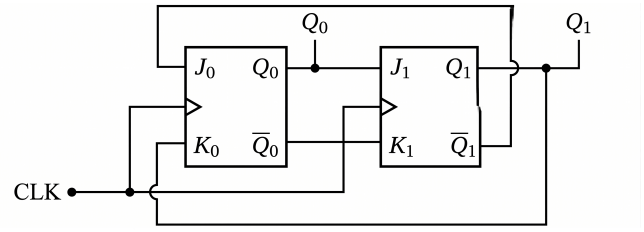



Fig. Q5.15.1

**A.** 00 **B.** 01
**C.** 10 **D.** 11

### Q5.15.2 NAT 2015 2M

The figure shows a digital circuit constructed using negative edge triggered $J$-$K$ flip flops. Assume a starting state of $Q_2\,Q_1\,Q_0 = 000$ will repeat after _____ number of cycles of the clock.

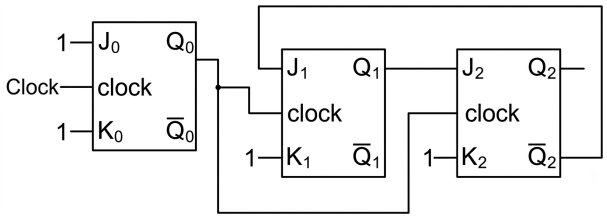



Fig. Q5.15.2

### Q5.14.1 MCQ 2014 1M

A state diagram of a logic which exhibits a delay in the output is shown in the figure, where $X$ is the don't care condition, and $Q$ is the output representing the state.

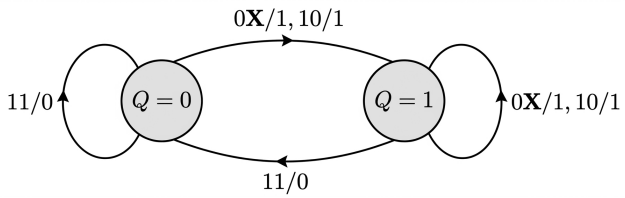



Fig. Q5.14.1

The logic gate represented by the state diagram is

**A.** XOR **B.** OR
**C.** AND **D.** NAND

**Q5.14.2** MCQ 2014 1M

Two monoshot multivibrators, one positive edge triggered ($M_1$) and another negative edge triggered ($M_2$) are connected as shown in figure. The monoshots $M_1$ and $M_2$ when triggered produce pulses of width $T_1$ and $T_2$ respectively, where $T_1 > T_2$. The steady state output voltage $V_0$ of the circuit is

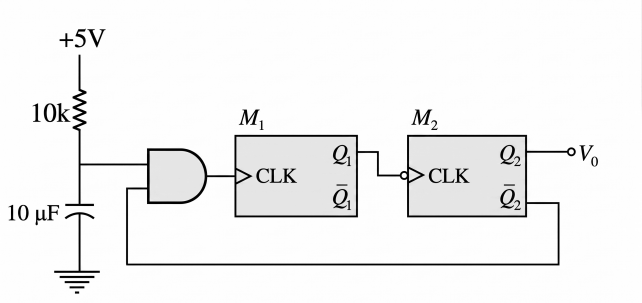
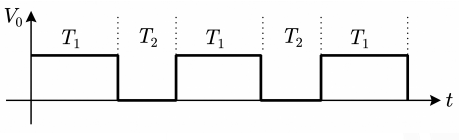
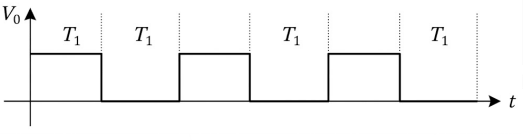
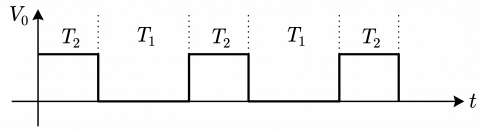
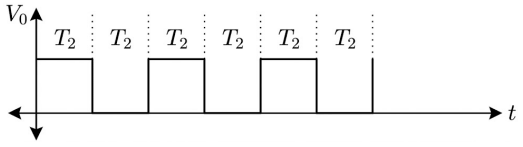



Fig. Q5.14.2

**A.**



**B.**



**C.**



**D.**


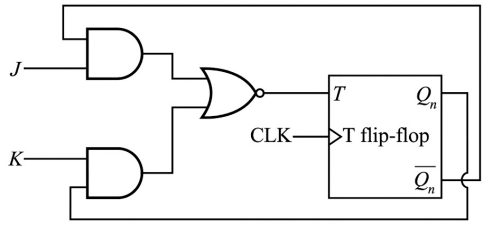

**Q5.14.3** MCQ 2014 1M

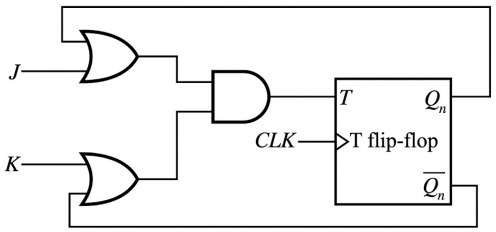
A cascade of three identical modulo-5 counters has an overall modulus of

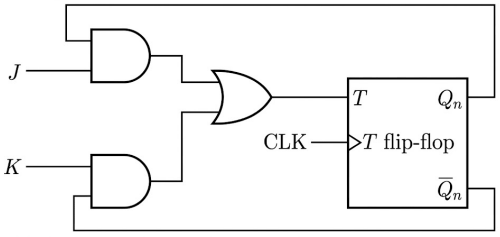
**A.** 5 **B.** 25
**C.** 125 **D.** 625

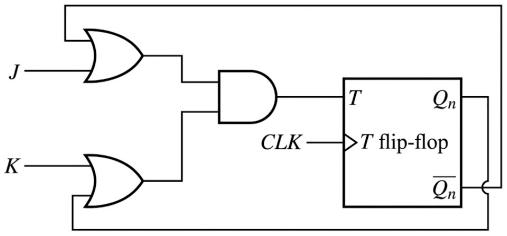
**Q5.14.4** MCQ 2014 2M

A $J$-$K$ flip-flop can be implemented by $T$ flip-flops. Identify the correct implementation.

**A.**



**B.**



**C.**



**D.**



**Q5.13.1** MCQ 2013 2M

The clock frequency applied to the digital circuit shown in figure below is 1 kHz. If the initial state of the output $Q$ of the flip-flop is '0', then the frequency of the output waveform $Q$ in kHz is

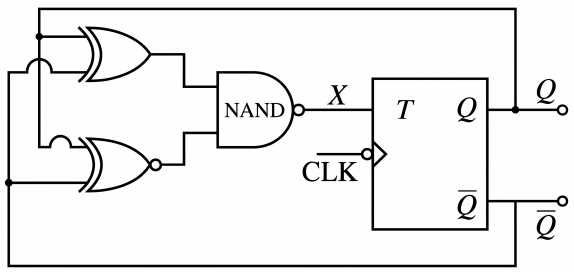



Fig. Q5.13.1

**A.** 0.25 **B.** 0.5
**C.** 1 **D.** 2

**Q5.12.1** MCQ 2012 1M

Consider the given circuit. In this circuit, the race around

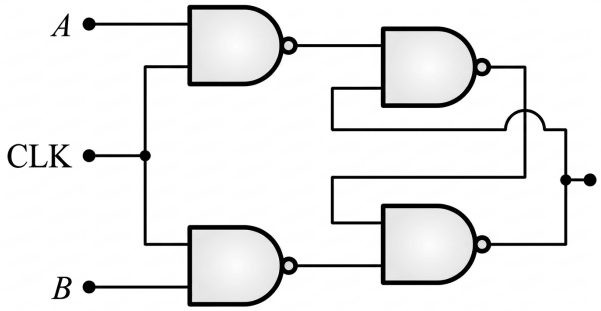



Fig. Q5.12.1

**A.** does not occur.

**B.** occurs when CLK = 0.

**C.** occurs when CLK = 1 and $A = B = 1$.

**D.** occurs when CLK = 1 and $A = B = 0$.

**Q5.12.2** MCQ 2012 2M

The state transition diagram for the logic circuit shown is

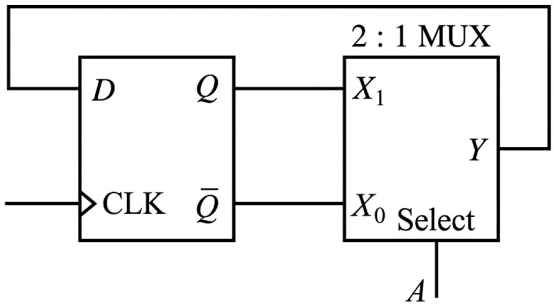
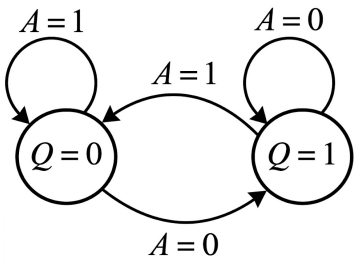
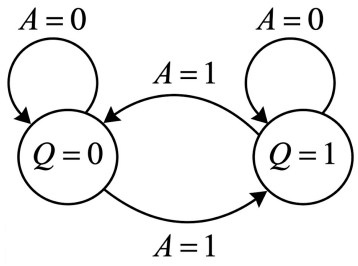


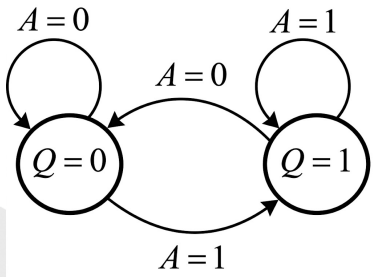

Fig. Q5.12.2



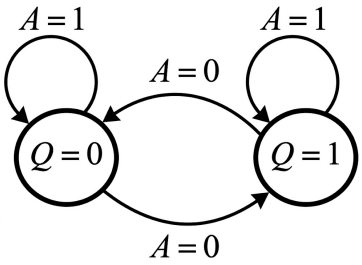

**A.**




**B.**




**C.**




**D.**

**Q5.11.1** MCQ 2011 2M

A 2-bit counter circuit is shown below. If the state $Q_A\,Q_B$ of the counter at the clock time $t_n$ is '10' then the state $Q_A\,Q_B$ of the counter at $t_n + 3$ (after three cycles) will be

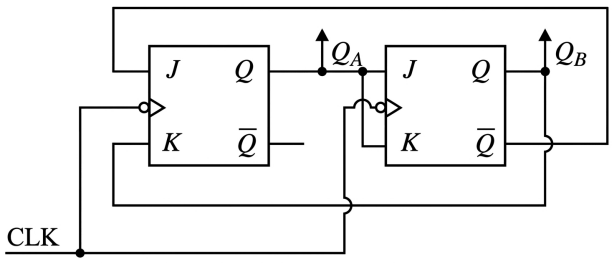



Fig. Q5.11.1

**A.** 00 **B.** 01
**C.** 10 **D.** 11

**Q5.08.1** MCQ 2008 1M

The truth table of a monoshot shown in the figure is given in the table below.

| X | Y | Q | $\bar{Q}$ |
|---|---|---|---|
| 0 | ↑ | ⊓ | ⊔ |
| ↓ | 1 | ⊓ | ⊔ |

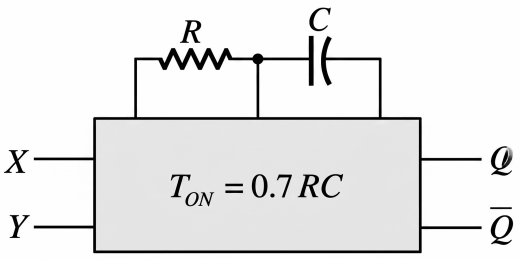



**Fig. Q5.08.1**

Two monoshots, one positive edge triggered and other negative edge triggered are connected as shown in the figure. The pulse widths of the two monoshot outputs $Q_1$ and $Q_2$ are $T_{ON_1}$ and $T_{ON_2}$ respectively.

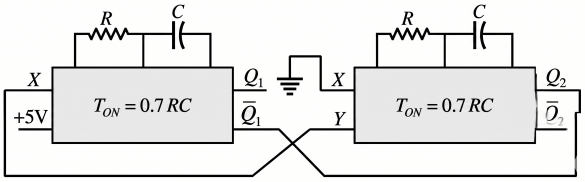



**Fig. Q5.08.1**

The frequency and the duty cycle of the signal at $Q_1$ will respectively be

**A.** $f = \dfrac{1}{T_{ON_1} + T_{ON_2}}, D = \dfrac{T_{ON_1}}{T_{ON_1} + T_{ON_2}}$

**B.** $f = \dfrac{1}{T_{ON_1} + T_{ON_2}}, D = \dfrac{T_{ON_2}}{T_{ON_1} + T_{ON_2}}$

**C.** $f = \dfrac{1}{T_{ON_1}}, D = \dfrac{T_{ON_1}}{T_{ON_1} + T_{ON_2}}$

**D.** $f = \dfrac{1}{T_{ON_2}}, D = \dfrac{T_{ON_1}}{T_{ON_1} + T_{ON_2}}$

**Q5.05.1** MCQ 2005 2M

The digital circuit shown in the figure works as

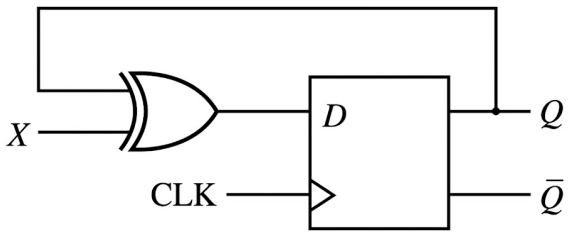



**Fig. Q5.05.1**

**A.** *J-K* flip-flop

**B.** Clocked *R-S* flip-flop

**C.** *T* flip-flop

**D.** Ring counter

**Q5.05.2** MCQ 2005 1M

Select the circuit which will produce the given output $Q$ for the input signals $X_1$ and $X_2$ given in the figure.

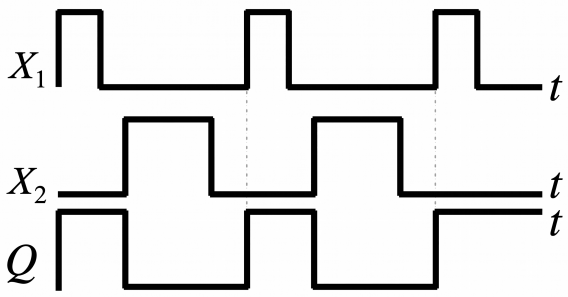
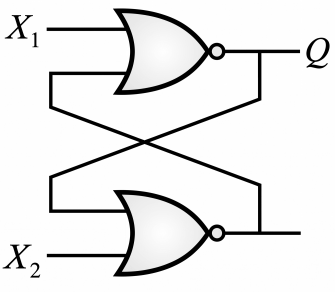
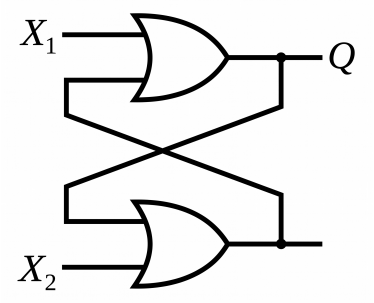



**Fig. Q5.05.2**

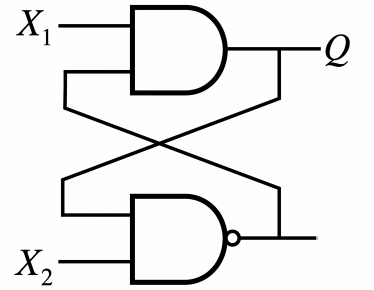
**A.**



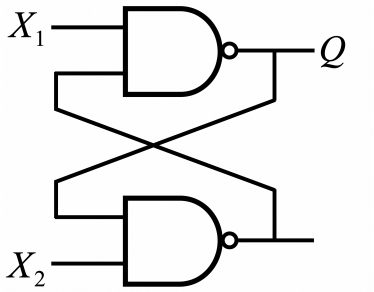
B.




C.




D.




**Q5.05.3** MCQ 2005 2M

In the figure, as long as $X_1 = 1$ and $X_2 = 1$, the output $Q$ remains

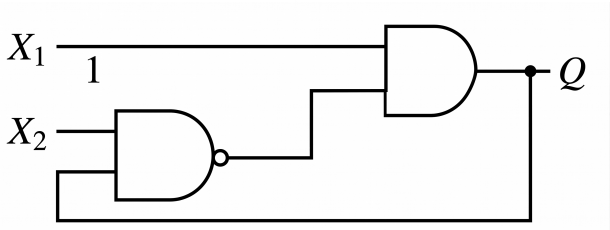



Fig. Q5.05.3

**A.** at 1

**B.** at 0

**C.** at its initial value

**D.** unstable

**Q5.04.1** MCQ 2004 1M

The digital circuit using two inverters shown in figure will act as

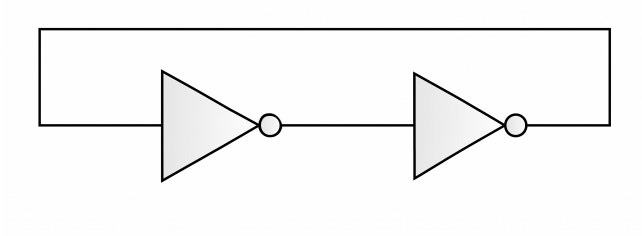


Fig. Q5.04.1

**A.** a bistable multivibrator

**B.** an astable multivibrator

**C.** a monostable multivibrator

**D.** an oscillator

**Q5.04.2** MCQ 2004 5M

The digital circuit shown in figure generates a modified clock pulse at the output. Choose the correct output waveform from the options given below.

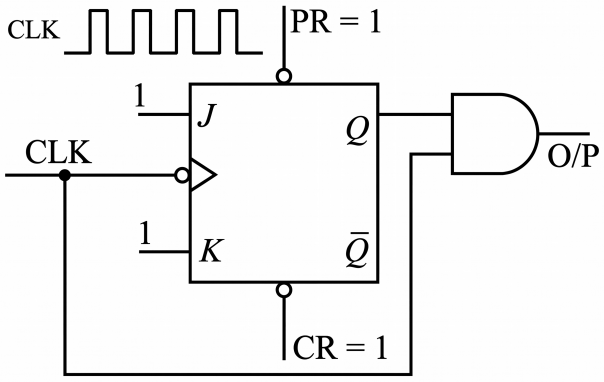



Fig. Q5.04.2

**A.**

**B.**

**C.**

**D.**

**Q5.03.1** MCQ 2003 2M

The shift register shown in figure is initially loaded with the bit pattern 1010. Subsequently the shift register is clocked and with each clock pulse the pattern gets shifted by one bit position to the right. With each shift, the bit at the serial input is pushed to the left most position (msb). After how many clock pulses will the content of the shift register become 1010 again?

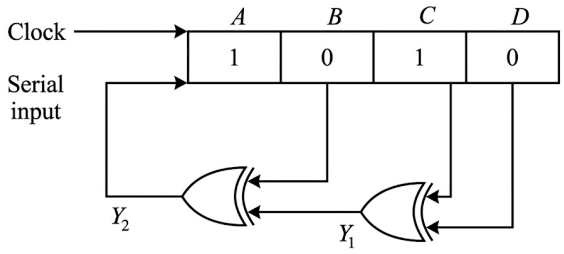



Fig. Q5.03.1

**A.** 3 **B.** 7
**C.** 11 **D.** 15

**Q5.03.2** MCQ 2003 1M

An $X$-$Y$ flip-flop, whose characteristic table is given below, is to be implemented using a $J$-$K$ flip-flop.

| $X$ | $Y$ | $Q_{n+1}$ |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | $Q_n$ |
| 1 | 0 | $\bar{Q}_n$ |
| 1 | 1 | 0 |

This can be done by making

**A.** $J = X,\ K = \bar{Y}$

**B.** $J = \bar{X},\ K = Y$

**C.** $J = Y,\ K = \bar{X}$

**D.** $J = \bar{Y},\ K = X$

**Q5.02.1** MCQ 2002 1M

The frequency of the clock signal applied to the rising edge triggered $D$ flip-flop shown in figure is 10 kHz. The frequency of the signal available at $Q$ is

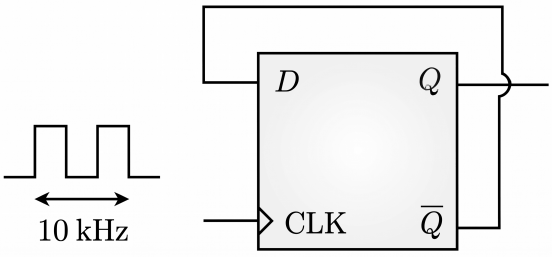



Fig. Q5.02.1

**A.** 10 kHz **B.** 2.5 kHz
**C.** 20 kHz **D.** 5 kHz

**Q5.02.2** MCQ 2002 5M

The ripple counter shown in figure is made up of negative edge triggered $J$-$K$ flip-flops. The signals levels at $J$ and $K$ inputs of all the flip-flops are maintained at logic 1. Assume all the outputs are cleared just prior to applying the clock signal. Module number of the counter is

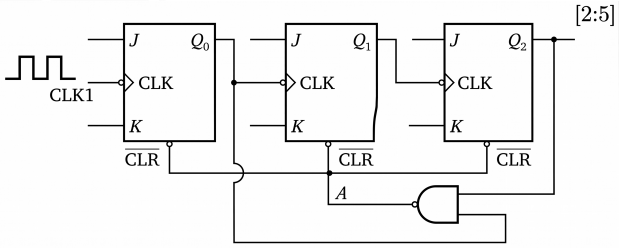



Fig. Q5.02.2

**A.** 7 **B.** 5
**C.** 4 **D.** 8

**Q5.01.1** MCQ 2001 5M

The circuit shown in the figure is a MOD-$N$ ring counter. Value of $N$ is (assume initial state of the counter is 1110 i.e. $Q_3\,Q_2\,Q_1\,Q_0 = 1110$)

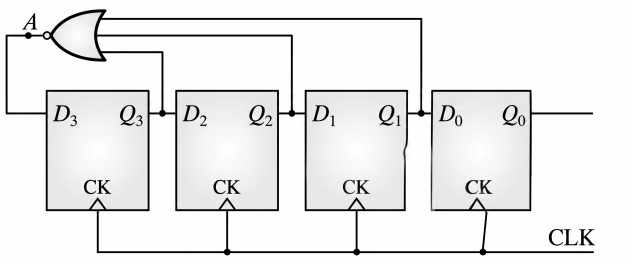



Fig. Q5.01.1

**A.** 4 **B.** 5
**C.** 7 **D.** 6

**Q5.99.1** MCQ 1999 1M

For a flip-flop formed using two NAND gates as shown in figure, the unstable state corresponds to

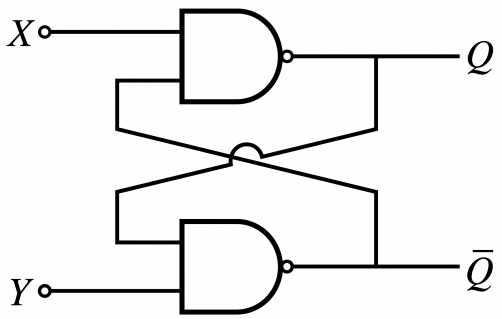



Fig. Q5.99.1

**A.** $X = 0, Y = 0$ **B.** $X = 0, Y = 1$
**C.** $X = 1, Y = 0$ **D.** $X = 1, Y = 1$

**Q5.95.1** NAT 1995 1M

For a $J$-$K$ flip-flop, its $J$ input is tied to its own $\bar{Q}$ output and its $K$ input is connected to its own $Q$ output. If the flip-flop is fed with a clock of frequency 1 MHz, its $Q$ output frequency will be _____ MHz.

**Q5.91.1** MCQ 1991 1M

The three stage Johnson ring counter shown in figure is clocked at a constant frequency of $f_C$ from the starting state of $Q_0 Q_1 Q_2 = 100$. The frequency of outputs $Q_0$, $Q_1$, $Q_2$ will be

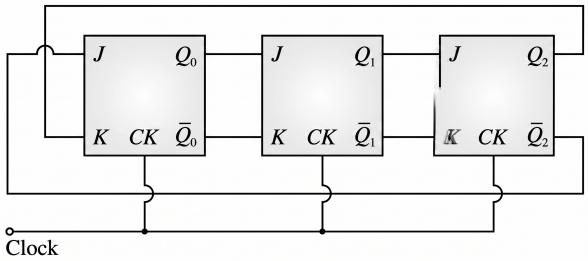



Fig. Q5.91.1

**A.** $\frac{f_C}{6}, \frac{f_C}{3}, \frac{f_C}{6}$

**B.** $\frac{f_C}{3}, \frac{f_C}{6}, \frac{f_C}{3}$

**C.** $\frac{f_C}{6}, \frac{f_C}{6}, \frac{f_C}{6}$

**D.** $\frac{f_C}{3}, \frac{f_C}{3}, \frac{f_C}{3}$

## Answer Key

| | | |
|---|---|---|
| Q5.24.1 C | Q5.23.1 8 | Q5.22.1 10 |
| Q5.22.2 12.5 | Q5.21.1 2.05 | Q5.21.2 C |
| Q5.20.1 C | Q5.18.1 B | Q5.17.1 6 |
| Q5.16.1 C | Q5.15.1 B | Q5.15.2 6 |
| Q5.14.1 D | Q5.14.2 C | Q5.14.3 C |
| Q5.14.4 B | Q5.13.1 B | Q5.12.1 A |
| Q5.12.2 D | Q5.11.1 C | Q5.08.1 A |
| Q5.05.1 C | Q5.05.2 A | Q5.05.3 D |
| Q5.04.1 A | Q5.04.2 B | Q5.03.1 B |
| Q5.03.2 D | Q5.02.1 D | Q5.02.2 B |
| Q5.01.1 A | Q5.99.1 A | Q5.95.1 0.5 |
| Q5.91.1 C | | |

UNIT

# VI

# ADC & DAC

## Topics

1. Introduction to DAC — Weighted Resistor and R-2R Ladder Type
2. Introduction to ADC — Counter Type
3. SAR Type ADC
4. Flash Type ADC
5. Single Slope and Dual Slope ADC
6. Sample and Hold Operation in ADC

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## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 11 | **MCQ:** 9 | **MSQ:** 0 | **NAT:** 2 |
| **1M:** 7 | **2M:** 4 | **Descriptive:** 0 | **Avg Weightage** ~2M | **Range:** 1M(2016)–1M(2016) |

**See preface for how Avg Weightage and Range are calculated.*

**Q6.16.1** MCQ 2016 2M

A 2-bit flash Analog to Digital Converter (ADC) is given below. The input is $0 \le V_{IN} \le 3$ V. The expression for the LSB of the output $B_0$ as a Boolean function of the inputs $X_2$, $X_1$, and $X_0$ is

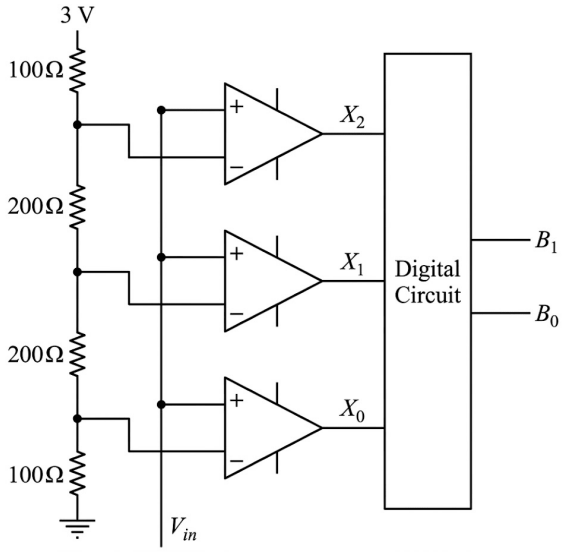


Fig. Q6.16.1

**A.** $X_0[\overline{X_2 \oplus X_1}]$

**B.** $\overline{X}_0[\overline{X_2 \oplus X_1}]$

**C.** $X_0[X_2 \oplus X_1]$

**D.** $\overline{X}_0[X_2 \oplus X_1]$

**Q6.15.1** MCQ 2015 2M

An 8-bit unipolar successive approximation register type ADC is used to convert 3.5 V to digital equivalent output. The reference voltage is +5 V. The output of ADC at the end of 3rd clock pulse after the start of conversion is

**A.** 1010 0000 **B.** 1000 0000
**C.** 0000 0001 **D.** 0000 0011

**Q6.06.1** MCQ 2006 2M

A student has made a 3-bit binary down counter and connected it to the R-2R ladder type DAC [Gain = $(-1\,\text{k}\Omega/2R)$] as shown in the figure to generate a staircase waveform. The output achieved is different as shown in the figure. What could be the possible cause of this error?

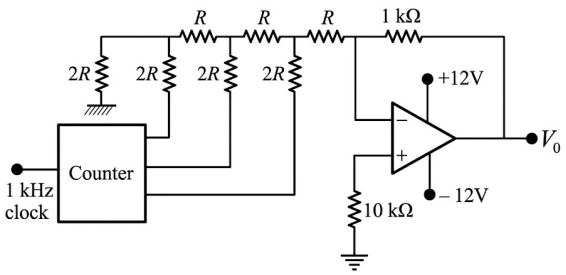


Fig. Q6.06.1

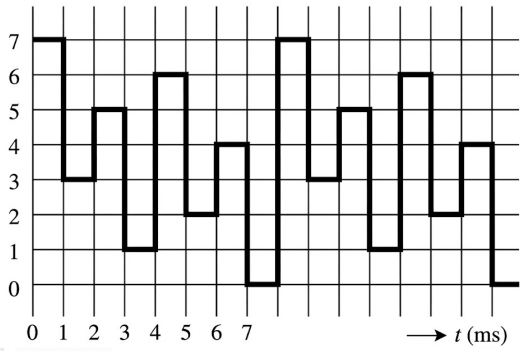


Fig. Q6.06.1

**A.** The resistance values are incorrect.

**B.** The counter is not working properly.

**C.** The connection from the counter to DAC is not proper.

**D.** The $R$ and $2R$ are interchanged.

**Q6.05.1** NAT 2005 2M

A digital-to-analog converter with a full-scale output voltage of 3.5 V has a resolution close to 14 mV. Its bit size is

**Q6.04.1** MCQ 2004 1M

The voltage comparator shown in figure can be used in the analog-to-digital conversion as

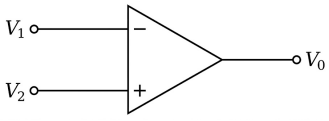


Fig. Q6.04.1

**A.** A 1-bit quantizer

**B.** A 4-bit quantizer

**C.** A 2-bit quantizer

**D.** A 8-bit quantizer

**Q6.01.1** MCQ 2001 1M

Among the following four, the slowest ADC (analog-to-digital converter) is

**A.** Parallel-comparator (i.e. flash) type

**B.** Successive approximation type

**C.** Integrating type

**D.** Counting type

**Q6.00.1** MCQ 2000 1M

A dual-slope analog-to-digital converter uses an $n$-bit counter. When the input signal $V_a$ is being integrated, the counter is allowed to count up to a value

**A.** Equal to $2^N - 2$

**B.** Equal to $2^N - 1$

**C.** Proportional to $V_a$

**D.** Inversely proportional to $V_a$

**Q6.94.1** MCQ 1994 1M

The number of comparisons carried out in a 4-bit flash type A/D converter is

**A.** 16 **B.** 15
**C.** 4 **D.** 3

**Q6.93.1** MCQ 1993 1M

A 10 bit A/D converter is used to digitize an analog signal in the 0 to 5 V range. The maximum peak to peak ripple voltage that can be allowed in the dc supply voltage is

**A.** Nearly 100 mV

**B.** Nearly 50 mV

**C.** Nearly 25 mV

**D.** Nearly 5 mV

**Q6.92.1** MCQ 1992 1M

The number of comparators needed in a parallel conversion type 8-bit A/D converter is

**A.** 8 **B.** 16
**C.** 255 **D.** 256

**Q6.91.1** NAT 1991 1M

The resolution of an 8-bit A/D converter is _____ %.

## Answer Key

| | | |
|---|---|---|
| Q6.16.1 **A** | Q6.15.1 **A** | Q6.06.1 **C** |
| Q6.05.1 **8** | Q6.04.1 **A** | Q6.01.1 **C** |
| Q6.00.1 **B** | Q6.94.1 **B** | Q6.93.1 **D** |
| Q6.92.1 **C** | Q6.91.1 **0.392** | |

UNIT

# VII

# Logic Families and Semiconductor Memories

## Topics

1. Logic Gate Implementation Using CMOS
2. Characteristics of Logic Families
3. Hazards in Combinational Circuits
4. Programmable Logic Devices (PLD, PLA, PAL)
5. ROM, SRAM and DRAM

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## Unit Snapshot*

| | | | | |
|---|---|---|---|---|
| **Stream:** EE | **Questions:** 8 | **MCQ:** 8 | **MSQ:** 0 | **NAT:** 0 |
| **1M:** 5 | **2M:** 3 | **Descriptive:** 0 | **Avg Weightage** ~2M | **Range:** 1M(2017)–2M(2026) |

**See preface for how Avg Weightage and Range are calculated.*

**Q7.26.1** MCQ 2026 2M

The MOSFET switches shown in the circuit are ideal.

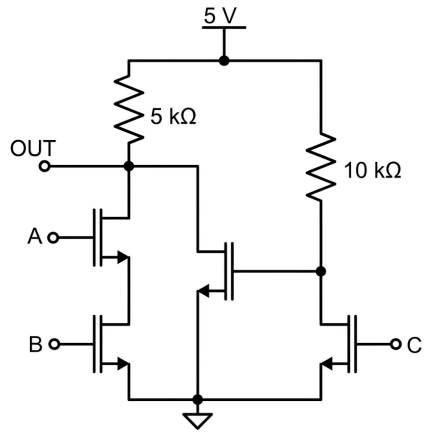



Fig. Q7.26.1

Which of the following is the correct option for Boolean logical expression of the output (OUT), and the maximum possible power ($P$) consumed by the circuit?

**A.** OUT = $\overline{AB + \overline{C}}$, $P = 5$ mW

**B.** OUT = $\overline{(A + B)\overline{C}}$, $P = 5$ mW

**C.** OUT = $\overline{AB\overline{C}} + \overline{C}$, $P = 7.5$ mW

**D.** OUT = $\overline{ABC}$, $P = 7.5$ mW

**Q7.17.1** MCQ 2017 1M

The logical gate implemented using the circuit shown below, where $V_1$ and $V_2$ are inputs (with 0 V as digital 0 and 5 V as digital 1) and $V_{out}$ is the output, is

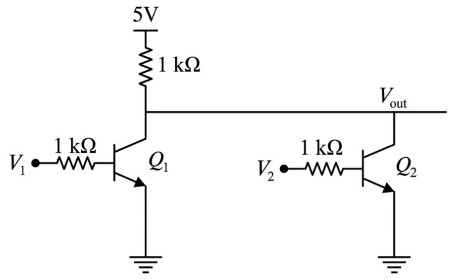



Fig. Q7.17.1

**A.** NOT **B.** NOR **C.** NAND **D.** XOR

**Q7.13.1** MCQ 2013 1M

In the circuit shown below, $Q_1$ has negligible collector-to-emitter saturation voltage and the diode drops negligible voltage across it under forward bias. If $V_{CC} = +5$ V, $X$ and $Y$ are digital signals with 0 V as logic 0 and $V_{CC}$ as logic 1, then the Boolean expression for $Z$ is

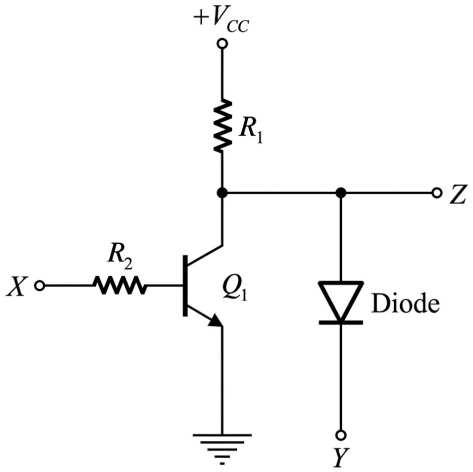



Fig. Q7.13.1

**A.** $XY$

**B.** $\bar{X}Y$

**C.** $X\bar{Y}$

**D.** $\overline{XY}$

**Q7.10.1** MCQ 2010 2M

The TTL circuit shown in the figure is fed with the waveform $X$ (also shown). All gates have equal propagation delay of 10 ns. The output $Y$ of the circuit is

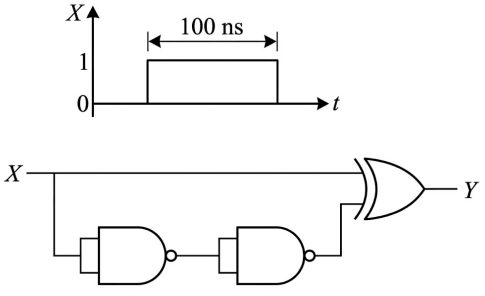
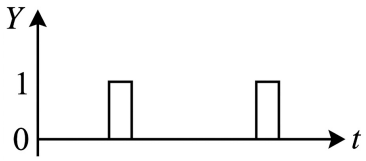
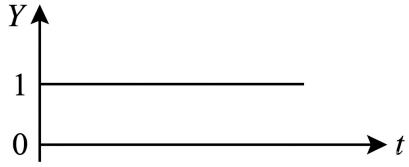



Fig. Q7.10.1

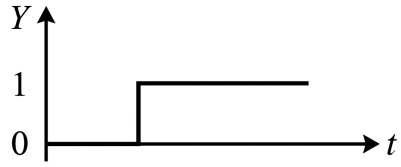
**A.**



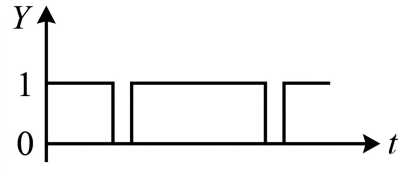

**B.**




**C.**




**D.**




## Q7.06.1 MCQ 2006 2M

A TTL NOT gate circuit is shown in figure. Assuming $V_{BE} = 0.7$ V of both the transistors, if $V_i = 3.0$ V, then the states of the two transistors are

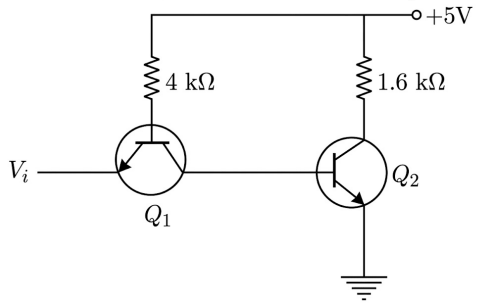



**Fig. Q7.06.1**

**A.** $Q_1$ ON and $Q_2$ OFF

**B.** $Q_1$ reverse ON and $Q_2$ OFF

**C.** $Q_1$ reverse ON and $Q_2$ ON

**D.** $Q_1$ OFF and $Q_2$ reverse ON

## Q7.05.1 MCQ 2005 1M

If $X_1$ and $X_2$ are the inputs to the circuit shown in the figure, the output $Q$ is

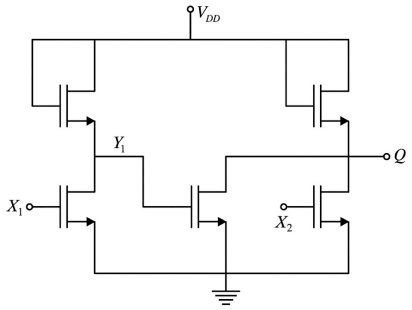



**Fig. Q7.05.1**

**A.** $\overline{X_1 + X_2}$

**B.** $\overline{X_1 \cdot X_2}$

**C.** $\overline{X_1} \cdot X_2$

**D.** $X_1 \cdot \overline{X_2}$

## Q7.98.1 MCQ 1998 1M

The open collector outputs of two 2-input NAND gates are connected to a common pull-up resistor. If the inputs to the gates are $P$, $Q$ and $R$, $S$ respectively, the output is equal to

**A.** $\overline{PQ} \cdot \overline{RS}$ **B.** $\overline{PQ} + \overline{RS}$

**C.** $PQ + RS$ **D.** $PQRS$

## Q7.98.2 MCQ 1998 1M

In standard TTL gates, the totem pole output stage is primarily used to

**A.** increase the noise margin of the gate.

**B.** decrease the output switching delay

**C.** facilitate a wired OR logic connection

**D.** increase the output impedance of the circuit.

## Answer Key

| | | |
|---|---|---|
| Q7.26.1<br>D | Q7.17.1<br>B | Q7.13.1<br>B |
| Q7.10.1<br>A | Q7.06.1<br>C | Q7.05.1<br>D |
| Q7.98.1<br>A | Q7.98.2<br>B | |

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